

Carcinoma of the Pancreas

The Big Picture

Pancreatic cancer = a quiet killer. 6th leading cancer death (UK), no screening test, and **>85% are unresectable at diagnosis**. Most are **ductal adenocarcinoma of the head**. The exam themes: **painless obstructive jaundice + Courvoisier**, CT to stage, **Whipple/PPPD** for the lucky resectable few, and palliation (stent the jaundice) for the rest.

Risk Factors

- Demographic: age **65–75**, male, black ethnicity.
- Lifestyle: **smoking** (the key modifiable one).
- Genetic/medical: family history (2 first-degree relatives → 18–57× risk), **BRCA2**, **hereditary pancreatitis (50–70×)**, chronic pancreatitis (5–15×), Lynch, Peutz-Jeghers, FAP, **diabetes mellitus**.

Pathology

- **>85% ductal adenocarcinoma**, mostly **head of gland**; infiltrates along nerve sheaths, lymphatics, vessels. Liver + peritoneal mets common.
- Cystic tumours: **serous cystadenoma** (older women, benign) vs **mucinous (MCN, IPMN)** = malignant potential.
- **Periampullary cancer** (ampulla/distal CBD) = ~■ of head-region tumours, **better prognosis**.

Clinical Features

- **Painless obstructive jaundice** = classic (head/ampullary tumour). + pruritus, dark urine, pale stool, steatorrhoea, nausea, epigastric discomfort.
- No jaundice → vague: anorexia, weight loss (often dismissed).
- **New diabetic >50** with upper-abdominal symptoms, or **unexplained pancreatitis** = red flag.
- **Body/tail tumours** grow silently → present advanced/unresectable.
- **Back pain** = retroperitoneal infiltration (worrying).
- **Courvoisier's law** — palpable gallbladder + jaundice = unlikely stones, suspect malignancy.

Investigation

- Bloods, **US** (is bile duct dilated?), then **contrast-enhanced CT** = preferred staging test.
- **Contraindications to resection**: hepatic/peritoneal mets, distant nodes, **encasement of SMA/hepatic/coeliac artery**. (Tumour size, duodenal/gastric/colon invasion, local nodes are NOT contraindications.)
- MRI/MRCP — "**double duct sign**." ERCP + stenting if cholangitis, diagnostic doubt, or delay to surgery (not mandatory if resectable).
- **EUS + FNA** for tissue (avoids peritoneal spillage); avoid percutaneous biopsy of resectable tumours. **CA19-9** baseline (recurrence marker). Diagnostic laparoscopy spares unnecessary laparotomy.

Management

- **>85% unresectable** at presentation.
- **Resectable head/ampulla** → **PPPD** (pylorus-preserving pancreatoduodenectomy): removes duodenum + pancreatic head + distal bile duct + local lymphadenectomy. **Whipple** (adds gastric antrum) reserved for FAP/duodenal encroachment.

- **Total pancreatectomy** — multifocal (main-duct IPMN) or unsafe anastomosis.
- **Body/tail** → **distal pancreatectomy + splenectomy** (vaccinate against pneumococcus/meningococcus/Hib).
- 3 phases (explore → resect → reconstruct), 3–6 h. **Mortality 3–5%, morbidity 30–40%**; pancreatic **anastomotic leak $\geq 10\%$** (octreotide reduces it).
- **Adjuvant** — 6 months gemcitabine $\pm$ 5-FU; chemoradiotherapy if R1 margin. Ampullary 5-yr survival 40%.

Palliation (for the 85%)

- Median survival: locally advanced 6–10 mo, metastatic 2–6 mo.
- **Jaundice** → ERCP stent (plastic cheaper/occludes; metal for longer life expectancy) or percutaneous transhepatic; surgical biliary bypass if found at laparotomy.
- **Duodenal obstruction (~15%)** → gastrojejunostomy (early) or duodenal stent (late).
- **Pain** → analgesia ladder + **coeliac plexus block**. Steatorrhoea → enzymes; treat diabetes.

Walk-Away Points

- **Painless obstructive jaundice + Courvoisier** = pancreatic head cancer until proven otherwise.
- **CT stages it; arterial encasement / mets = unresectable.**
- **PPPD** is the standard curative op (mortality 3–5%, leak the feared complication).
- New diabetes >50 or unexplained pancreatitis → image the pancreas.
- Most present too late → palliate (stent jaundice, relieve duodenal obstruction, block pain).

Carcinoma of the Pancreas — Revision Table	
Feature	Detail
Commonest type	Ductal adenocarcinoma (>85%) , head of gland
Top modifiable risk	Smoking
Genetic risk	BRCA2, hereditary pancreatitis (50–70×), Lynch, Peutz-Jeghers, FAP
Hallmark presentation	Painless obstructive jaundice
Courvoisier's law	Palpable GB + jaundice → malignancy not stones
Red flags	New diabetic >50, unexplained pancreatitis, back pain
Cystic tumour	Behaviour
Serous cystadenoma	Older women, benign
Mucinous (MCN, IPMN)	Malignant potential
Periampullary	■ of head tumours, better prognosis
Investigation	Role
US	Bile duct dilated?
CT (contrast)	Staging test of choice
MRCP	"Double duct sign"
ERCP + stent	Cholangitis, doubt, delay to surgery
EUS + FNA	Tissue dx (no peritoneal spillage)
CA19-9	Baseline / recurrence marker
Unresectable if	Resectable despite
Liver/peritoneal mets	Tumour size
Distant nodes	Duodenum/stomach/colon invasion
SMA/hepatic/coeliac encasement	Local field nodes
Surgery	Indication
PPPD	Head/ampulla (standard)
Whipple (+antrum)	FAP, duodenal encroachment
Total pancreatectomy	Multifocal / main-duct IPMN
Distal pancreatectomy + splenectomy	Body/tail
Outcomes	Value
Resectable at dx	<15%
PPPD mortality	3–5%
Morbidity	30–40%
Pancreatic leak	≥10% (↓ with octreotide)
Adjuvant	Gemcitabine ± 5-FU 6 mo
Palliation	Method
Jaundice	ERCP stent / percutaneous / bypass
Duodenal obstruction (15%)	Gastrojejunostomy / duodenal stent
Pain	Analgesia + coeliac plexus block

Bold = high-yield.